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Editors

Journal of Chemical Information and Modeling

Dear Editors:

Please consider our Article, “Conditioning Protein Generation via Hopfield Pattern Multiplicity,” for publication in the *Journal of Chemical Information and Modeling*.

This work presents a training-free method for conditioning protein sequence generation when only a small designated subset of a family is available. Pattern multiplicities enter the modern Hopfield energy as logit biases, giving an exact Gaussian-mixture equilibrium target without retraining or changing the sampler’s asymptotic cost. The paper also distinguishes this latent target from the sequence-level distribution obtained after PCA reconstruction and discrete decoding, exposing and quantifying a practically important calibration gap.

The computational study includes five Pfam families and a curated ω -conotoxin family, replicated multiplicity sweeps, exact-equilibrium controls, hard-mask and hard-curation comparisons, and sequence, structure, and language-model assessments. We additionally introduce a matched HMMER3 benchmark that uses the same cleaned alignments, designation labels, multiplicity ratios, replicate counts, and output sizes. The comparison gives an informative tradeoff rather than a blanket superiority claim: the weighted profile HMM more directly transfers a selected position-specific marginal, whereas stochastic attention gives lower ESM2 pseudo-perplexity in the matched Kunitz comparison.

The manuscript fits *JCIM* as a new molecular-modeling and chemical-informatics methodology with explicit theory, reproducible computational benchmarks, and an honest account of its domain of validity. It is not submitted as a docking or experimental binding study. The low-confidence Cav2.2 complex predictions are retained in the Supporting Information only as a negative diagnostic and are not used as evidence of binding.

The manuscript and Supporting Information provide a Data and Software Availability statement. All curated inputs, canonical per-replicate outputs, benchmark emissions, scoring results, and reproduction scripts are provided in the public study repository. The author declares no competing financial interest.

A preprint is available as arXiv:2603.20115 (doi:10.48550/arXiv.2603.20115). This manuscript is not under consideration elsewhere.

Thank you for your consideration.

Sincerely,

Jeffrey D. Varner